PROJECT
IT9201

AI-DRIVEN PREDICTION AND ANALYSIS OF GENDER-BASED DISCRIMINATION IN BAHRAIN

Prepared by
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Semester B 2025 - 2026
Machine Learning and Data mining

TASK 1

PROBLEM AND OBJECTIVES-BACKGROUND OF THE STUDY

1.1 Background

Gender-based discrimination remains a challenge in workplaces despite global progress and reforms. Factors such as unequal leadership opportunities, limited mentorship, work-life balance issues, and inadequate institutional support contribute to these experiences. In Bahrain, addressing these issues supports national goals aligned with Bahrain Vision 2030 and relevant Sustainable Development Goals (SDGs 4, 5, 8, and 10).

Machine learning and AI offer techniques to uncover complex patterns in social data, supporting predictive analysis and explainable insights for policymakers, human resource departments and institutional leaders.

1.2 Problem Statement

Traditional statistical methods often fail to capture complex, nonlinear relationships among factors influencing discrimination. This project aims to develop and evaluate machine learning models that predict gender-based discrimination using demographic, workplace, and institutional data collected from surveys in Bahrain.

The research question is: Can workplace and institutional factors predict whether individuals in Bahrain have experienced or observed gender-based discrimination?

The target variable, experienced_discrimination, is a binary outcome based on self-reported responses (Yes = 1, No = 0).

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PROBLEM AND OBJECTIVES-BACKGROUND OF THE STUDY

1.3 Literature Review

The use of supervised machine learning in gender inequality research has grown significantly, though studies highlight the risk of bias in predictive models. Mehrabi et al. (2021) notes that models trained on biased historical data may reinforce inequalities, while also identifying relevant bias types in survey-based data. Ensemble models have also been used to predict the Gender Inequality Index (GII), showing strong performance but lacking the detail needed to capture individual and institutional dynamics, an area addressed in this study.

In association rule mining, Jacob, Raj, and Cheriyan (2025) find that structural and cultural factors, such as legal protections and attitudes toward gender-based violence, are stronger predictors than education. This aligns with this project, where perception-based features like gender_career_influence are more influential. By using a larger dataset and lower support threshold, this study generates more detailed patterns.

Research on AI adoption highlights gender gaps, including a reported productivity gap choosing male researchers due to differences in institutional support. Lütz (2022) adds that policy frameworks often lag behind technological advances, which increases bias risks, reflected in work-life balance and mentorship variables.

Moreover, Casad et al. (2020) identify structural barriers in STEM, such as unequal workload and limited mentorship, which are captured in this dataset. Clavero and Galligan (2021) emphasize accountability in Gender Equality Plans, supporting data-driven assessment, while Balima (2022) highlights the role of legal rights and workplace policies in shaping gender inequality in the MENA region.

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PROBLEM AND OBJECTIVES- BACKGROUND OF THE STUDY

1.4 Objectives

AIM

- The **dataset** should be cleaned, normalzied, encoded and split into training & testing sets.
- **Merge two survey based datasets** to maximize sample size and feature diversity.
- **Train and compare five supervised classifiers** (Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost)
- Apply **SMOTE** for class imbalance and compare performance with and without it.
- **Apply unsupervised techniques** (K-means clustering & Association Rule Mining)
- Perform **Hyperparameter tuning using GridSearchCV** and identify best model.

TASK 1

PROBLEM AND OBJECTIVES-BACKGROUND OF THE STUDY

1.5 Software Libraries & Tools

The project was implemented using Python in Google Colab using Python 3.12. The following libraries were used:

Library	Version	Purpose
pandas	2.2.2	Data loading, manipulation, merging
numpy	2.0.2	Numerical operations, array handling
scikit-learn	1.4+	ML models, preprocessing, evaluation
xgboost	2.0+	XGBoost classifier
imbalanced-learn	0.12+	SMOTE for class imbalance
mlxtend	0.23+	Apriori algorithm for ARM
matplotlib / seaborn	3.8+ / 0.13+	Visualizations
Orange (no-code)	3.36+	ARM and K-Means validation

Table 1: Software Libraries

TASK 2

PROPOSED ML FRAMEWORK WITH DATA ACQUISITION/PREPROCESS

2.1 Dataset Description

The project used two survey-based datasets related to gender equality in workplace and academic contexts in Bahrain. The datasets contained demographic information, workplace support indicators, mentorship access, leadership experience, work-life balance satisfaction, perceived bias, and discrimination experiences.

Dataset 1, "Survey on Gender Equality in Industries Across Bahrain," collected ~102 responses, while Dataset 2, "Survey on Gender Equality in Academia Across Bahrain," gathered ~62 responses. Due to the limited number of records in both datasets, additional data were generated using GenAI. Dataset 1 was expanded to 2,964 rows, and Dataset 2 to 2,062 rows through AI-generated responses.

Dataset 1 is moderately imbalanced, with 61.1% "No" and 38.9% "Yes" responses; therefore, SMOTE was applied to mitigate bias toward the majority class. In contrast, Dataset 2 is nearly balanced, with 52.9% "Yes" and 47.1% "No," making it a useful addition when combined, resulting in an overall class distribution of 54.4% "No" and 45.6% "Yes."

The distribution of missing values and class balance for both datasets is illustrated in Figure 2.1. To see more, please refer to Appendix XYZ. The target variable used for supervised learning was 'experienced_discrimination', where Yes = 1 and No = 0.

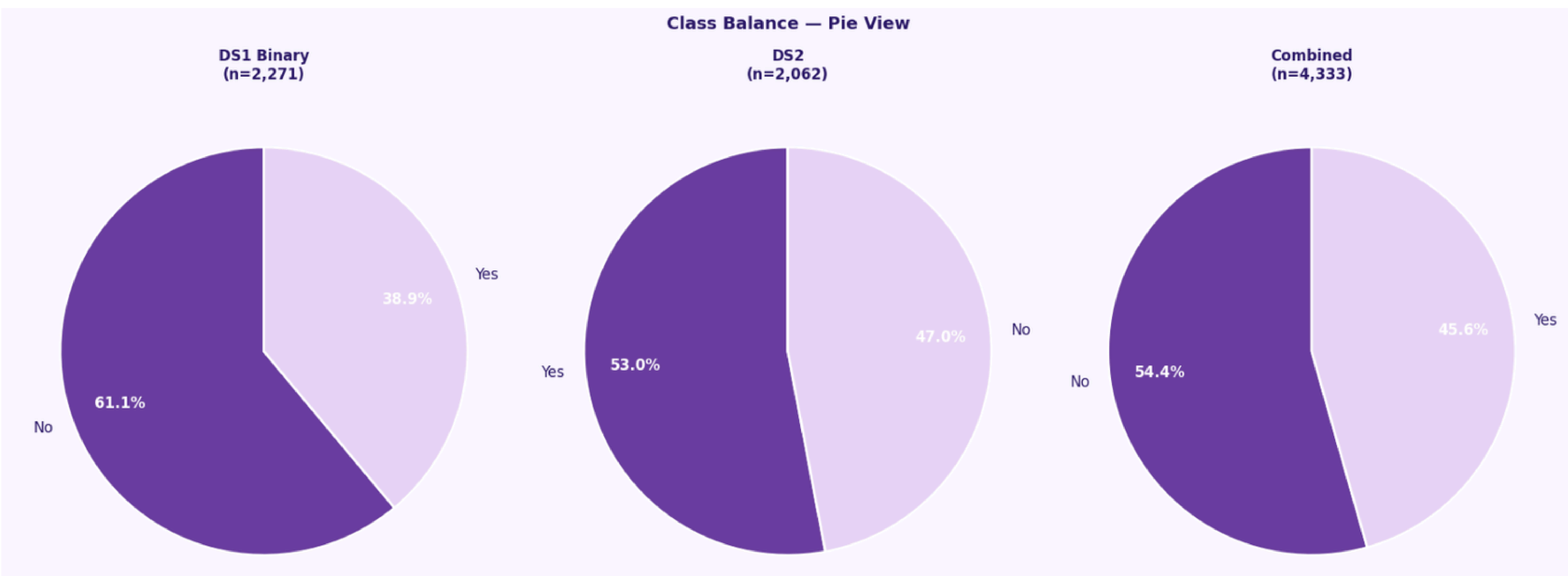

Figure 2.1: Pie charts showing class imbalance.

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Table 1: Shows the libraries used.

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PROPOSED ML FRAMEWORK WITH DATA ACQUISITION/PREPROCESS

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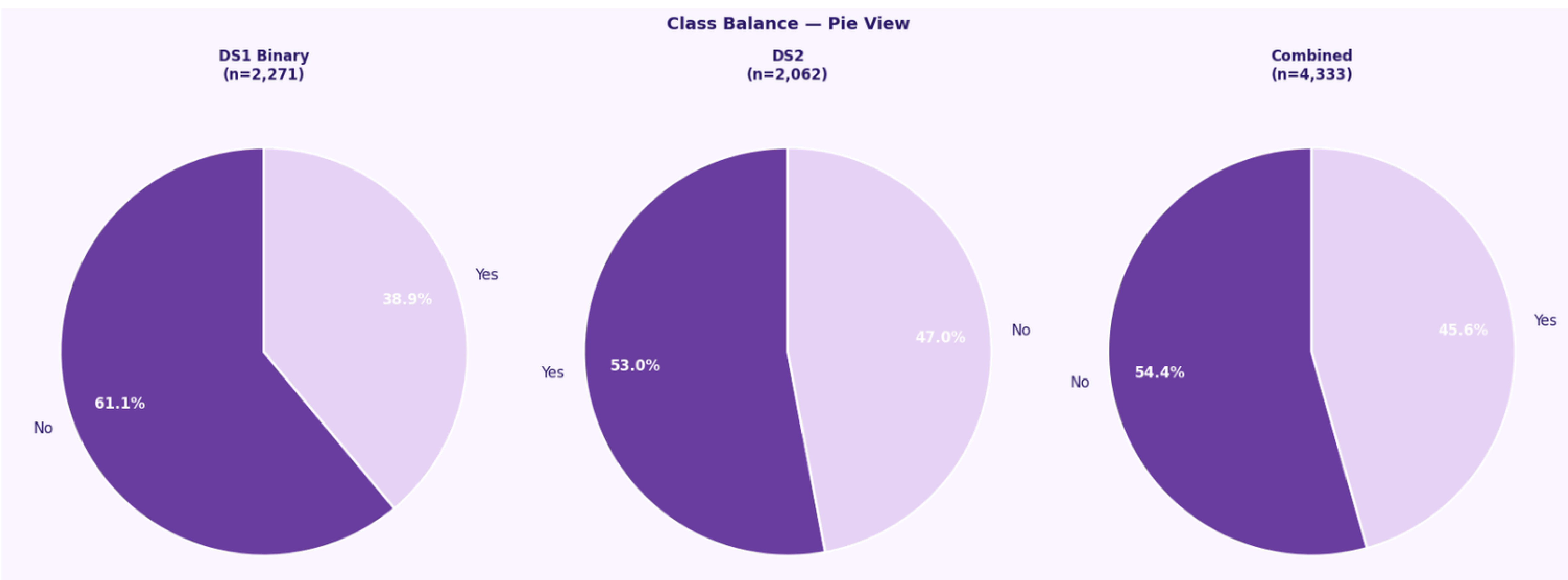

Figure 2.1: Pie charts showing class imbalance.

TASK 2

PROPOSED ML FRAMEWORK WITH DATA ACQUISITION/PREPROCESS

2.1 Dataset Description

<table>
<tr><th>Property</th><th>Dataset 1 — Industry Survey</th><th>Dataset 2 — Academia Survey</th></tr>
<tr><td>Full Name</td><td>Survey on Gender Equality in Industries Across Bahrain</td><td>Survey on Gender Equality in Academia Across Bahrain (AutoML Augmented)</td></tr>
<tr><td>Rows (raw)</td><td>2,964</td><td>2,062</td></tr>
<tr><td>Columns (raw)</td><td>29</td><td>37</td></tr>
<tr><td>Target Column Name</td><td>"Have you personally experienced or witnessed gender-based discrimination in your career?"</td><td>"Have you experienced or witnessed gender-based discrimination at any point in your academic career?"</td></tr>
<tr><td>Response Language</td><td>Bilingual — English + Arabic in same cell (e.g. "Yes نعم")</td><td>English only</td></tr>
<tr><td>Shared Core Features</td><td colspan="2">gender, age, employment_status, industry, field_of_study, leadership_role, mentorship_access, conference_speaker, awards_received, parental_leave_policy, parental_leave_taken, work_life_satisfaction, perceived_gender_bias, gender_career_influence, workload_differs, work_life_challenges, aware_of_rights, grievance_mechanism, health_counselor_access</td></tr>
<tr><td>Dataset 1-Only Feature</td><td>collab_frequency</td><td>—</td></tr>
<tr><td>Dataset 2-Only Features</td><td>—</td><td>nationality, research_active, career_break_pressure, family_study_delay, violence_justified_score, male_dominance_score, underreporting_stigma, religious_norms_equality, cultural_barriers, witnessed_violence, career_advice_focus_family, computer_literacy,</td></tr>
<tr><td>Missing Values</td><td>Minimal, <2% per shared column</td><td>None in target; some in shared columns</td></tr>
</table>

Table 2: Dataset Description.

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PROPOSED ML FRAMEWORK WITH DATA ACQUISITION/PREPROCESS

2.2 Dataset Processing Pipeline

01

Load Raw Dataset

Load both Excel files using pandas.read_excel. Verify shape and target distributions; DS1 contains bilingual labels, DS2 is clean English.

```
Dataset 1 shape: (2964, 29)
  Target (raw): {'No 1378 :'لا, 'Yes 879 :'نعم, 'Prefer not to say 687 :'أفضل عدم الإجابة, 'No\xa09 :'لا, 'Prefer not to say\xa05 :'الإجابة

Dataset 2 shape: (2062, 37)
  Target (raw): {'Yes': 1092, 'No': 970}

✓ Both datasets loaded successfully
  Combined raw rows: 5,026
```

02

Cleaning Datasets

- DS1: Rename columns by index, drop 8 metadata fields, and normalise 17 columns (e.g., all “Yes نعم” variants → “Yes”). Add DS2-only columns as NaN.
- DS2: Align column names with DS1. Standardise values: Gender (Woman→Female), bin Age into 5 groups, map Likert scores to text labels, and simplify employment to Employed. Add DS1-only column (collab_frequency) as NaN.

```
Dataset 1 cleaned: (2964, 35)
  Target: {'No': 1387, 'Yes': 884, 'Prefer not to say': 692}
  DS1-specific feature: collab_frequency has 2963 non-null values
```

```
Dataset 2 cleaned: (2062, 42)
  Target: {'Yes': 1092, 'No': 970}
  DS2-specific features (15+): nationality, research_active, violence_justified_score, etc.
  Age range in DS2: 23-71 → binned successfully
```

03

Merge Datasets

Concatenate datasets using pd.concat (because no common IDs). Remove “Prefer not to say” targets to produce a clean binary classification problem. Final dataset: 4,333 rows, 35 columns, 54.4% No / 45.6% Yes.

```
Class balance: Yes=45.6%  No=54.4%
→ Dataset is much more balanced vs single-dataset baseline (was 31% / 69%)
```

TASK 2

PROPOSED ML FRAMEWORK WITH DATA ACQUISITION/PREPROCESS

2.2 Dataset Processing Pipeline

04

Imputation and EDA

Fill structural and minor missing values using mode. Group rare categories as “Other”. Generate 6 EDA charts (target, gender, age, dataset, satisfaction, career influence).

```
Missing values before imputation:
                            Missing  Missing %
industry                         12        0.3
field_of_study                    1        0.0
leadership_role                   1        0.0
collab_frequency               2062       47.6
parental_leave_taken           2062       47.6
aware_of_rights                   1        0.0
nationality                    2271       52.4
research_active                2271       52.4
career_break_pressure          4333      100.0
family_study_delay             4333      100.0
violence_justified_score       4333      100.0
male_dominance_score           2271       52.4
underreporting_stigma          2271       52.4
religious_norms_equality       2271       52.4
cultural_barriers              2271       52.4
witnessed_violence             2271       52.4
career_advice_focus_family     2271       52.4
computer_literacy              2271       52.4
overall_gender_equality_score  2271       52.4

Missing after imputation: 12999 ✓
Rows: 4333
```

05

Feature Encoding

- OrdinalEncoder (14 cols): ordered features (e.g., satisfaction)
- LabelEncoder (10 cols): binary (Yes/No → 0/1)
- One-Hot (6 cols → 52 dummies): nominal categories (no ranking)

```
Encoded dataset shape: (4333, 77)
Total features: 76
Target distribution: {0: 2357, 1: 1976}

Encoding summary:
  Ordinal encoded: 14 columns
  Binary encoded:  10 columns
  One-hot encoded: 6 columns → expanded to 52 dummies

New DS2-sourced encoded features:
  research_active (ordinal), underreporting_stigma (binary), career_break_pressure (binary)
  violence_justified_score (numeric), male_dominance_score (numeric)
  nationality (one-hot), religious_norms_equality (one-hot), cultural_barriers (one-hot)
```

TASK 2

PROPOSED ML FRAMEWORK WITH DATA ACQUISITION/PREPROCESS

2.2 Dataset Processing Pipeline

06

Train & Test Split

The dataset was split into: 80% training and 20% testing using stratified sampling to preserve class balance.

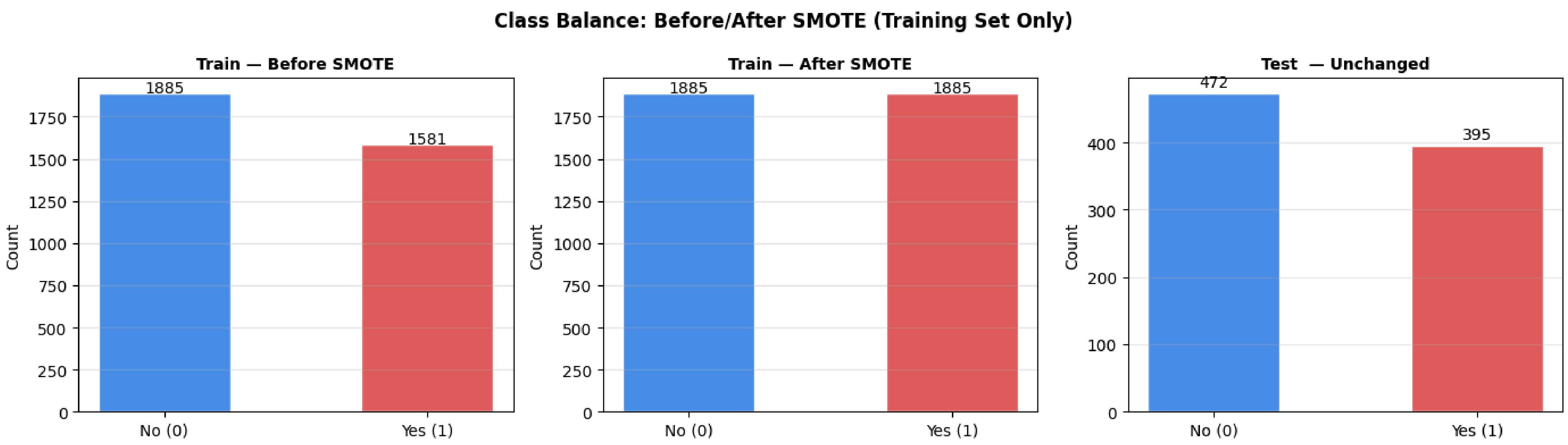
```
Train: 3466 rows | Test: 867 rows
Train class: {0: 1885, 1: 1581}
Test class:  {0: 472, 1: 395}
```

07

Scaling & SMOTE

StandardScaler was applied to normalize feature ranges. SMOTE was then applied only to the training set to balance the minority class and reduce prediction bias toward the majority class.

```
SMOTE effect:
  Before: {np.int64(0): np.int64(1885), np.int64(1): np.int64(1581)}
  After:  {np.int64(0): np.int64(1885), np.int64(1): np.int64(1885)}
  Synthetic samples added: 304
```



TASK 3

CHOICE OF FEATURE SELECTION, LEARNING METHODS, EVALUATION STRATEGIES AND METRICS

3.1 Supervised Models

The objective of this study is to identify suitable machine learning and data mining techniques for predicting gender-based discrimination. The task is formulated as a binary classification problem, where each model predicts whether a respondent has experienced or observed discrimination (Refer to Appendix 7.3).

Five supervised learning models were implemented under two conditions: with and without SMOTE. Each algorithm was selected to represent a specific role, ranging from interpretable models to advanced ensemble techniques:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost

```python
# ── CELL 9: Train All Classifiers ──────────────────────────────────────

MODELS = {
    'Logistic Regression': LogisticRegression(max_iter=1000, random_state=42),
    'Decision Tree':        DecisionTreeClassifier(random_state=42),
    'Random Forest':        RandomForestClassifier(n_estimators=100, random_state=42),
    'Gradient Boosting':    GradientBoostingClassifier(n_estimators=100, random_state=42),
    'XGBoost':              XGBClassifier(n_estimators=100, random_state=42,
                                          eval_metric='logloss', verbosity=0),
}

results        = []
trained_models = {}

print('Training 10 model variants (5 models × 2 SMOTE conditions)...')
print()

for name, model in MODELS.items():
    for smote_label, Xtr, ytr in [('No SMOTE',   X_train_sc, y_train),
                                  ('With SMOTE', X_train_sm, y_train_sm)]:
        m = copy.deepcopy(model)
        m.fit(Xtr, ytr)
        yp  = m.predict(X_test_sc)
        ypp = m.predict_proba(X_test_sc)[:, 1]

        row = {
            'Model':     name,
            'SMOTE':     smote_label,
            'Accuracy':  round(accuracy_score(y_test,  yp) * 100, 2),
            'Precision': round(precision_score(y_test, yp) * 100, 2),
            'Recall':    round(recall_score(y_test,    yp) * 100, 2),
            'F1-Score':  round(f1_score(y_test,        yp) * 100, 2),
            'ROC-AUC':   round(roc_auc_score(y_test,  ypp) * 100, 2),
        }
        results.append(row)
        trained_models[f'{name}_{smote_label}'] = m
        print(f'  [{smote_label:10}] {name:25}  '
              f'Acc={row["Accuracy"]:5.2f}%  F1={row["F1-Score"]:5.2f}%  AUC={row["ROC-AUC"]:5.2f}%')

df_results = pd.DataFrame(results)
print(f'\n✓ {len(trained_models)} model variants trained')
```

Figure 1: Code for supervised learning

TASK 3

CHOICE OF FEATURE SELECTION, LEARNING METHODS, EVALUATION STRATEGIES AND METRICS

Model	Base Parameters	Why These Parameters?	Justification/Role
Logistic Regression	max_iter=1000, C=1.0, penalty=l2, solver=lbfgs	max_iter=1000 ensures convergence on 76 features without exceeding iteration limits. Default C=1.0 applies moderate L2 regularisation. Tuned to C=1, penalty=l2, solver=liblinear via GridSearchCV	Serves as an interpretable baseline, where coefficients clearly show the direction and strength of each feature's impact on discrimination probability.
Decision Tree	random_state=42, no depth limit (base); max_depth=3 for explainability demo	No depth restriction is applied during training to fully capture patterns; however, a depth-3 version is presented for readability. Deep trees risk overfitting without pruning.	Functions as an explainability model, as its decision-making process can be directly interpreted through if/then rules.
Random Forest	n_estimators=100, max_features='sqrt', random_state=42	Using 100 trees balances accuracy and efficiency, while sqrt feature selection reduces correlation among trees. Tuned parameters include n_estimators=200, max_depth=20, max_features='log2'.	Mitigates overfitting of a single Decision Tree by aggregating multiple trees; also provides reliable feature importance rankings.
Gradient Boosting	n_estimators=100, learning_rate=0.1, max_depth=3, random_state=42	A learning_rate=0.1 offers a balance between stability and convergence speed. Tuned with learning_rate=0.1, max_depth=5, subsample=1.0, and n_estimators=100.	Achieves the best performance by sequentially correcting previous errors and capturing remaining patterns.
XGBoost	n_estimators=100, eval_metric='logloss', verbosity=0, random_state=42	eval_metric='logloss' suits binary probability outputs. Built-in L1 (alpha) and L2 (lambda) regularisation reduce overfitting without extensive tuning.	Acts as an industry-standard benchmark, offering advanced regularisation and handling of missing values beyond standard Gradient Boosting.

Table 3: Models and their parameters.

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CHOICE OF FEATURE SELECTION, LEARNING METHODS, EVALUATION STRATEGIES AND METRICS

3.2 Why SMOT?

The merged dataset has a slight class imbalance (54% No, 46% Yes). Training without SMOTE shows the original distribution, while using SMOTE tests whether balancing improves detection of the minority class.
Results indicate that ensemble models (Random Forest, Gradient Boosting, XGBoost) perform better without SMOTE, as it fundamentally handles mild imbalance, whereas Logistic Regression benefits from SMOTE, showing improved recall through class balancing.

3.3 Evaluation Metrics

Metric	Formula	Why Used Here
Accuracy	(TP+TN) / Total	Measures overall prediction correctness; included for reference but not the main metric due to class imbalance
Precision	TP / (TP+FP)	Indicates how many predicted “Yes” cases are correct; important when false positives are costly.
Recall	TP / (TP+FN)	Measures how many actual “Yes” cases are correctly identified; crucial for detecting discrimination
F1-Score	2 × (Precision × Recall) / (Precision + Recall)	Balances Precision and Recall; key metric for imbalanced datasets
ROC-AUC	Area under ROC curve	Evaluates model performance across all thresholds; provides a comprehensive measure
Confusion Matrix	TP, FP, TN, FN breakdown	Displays error types; used to visualise model performance with SMOTE

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CHOICE OF FEATURE SELECTION, LEARNING METHODS, EVALUATION STRATEGIES AND METRICS

```
================================================================================
FULL MODEL PERFORMANCE — MERGED DATASET (n=4,333 | 76 features)
================================================================================
              Model      SMOTE  Accuracy  Precision  Recall  F1-Score  ROC-AUC
Logistic Regression    No SMOTE    78.32      78.36   72.41     75.26    85.65
Logistic Regression  With SMOTE    77.39      75.32   74.94     75.13    85.93
      Decision Tree    No SMOTE    75.20      71.53   75.70     73.55    75.24
      Decision Tree  With SMOTE    75.43      73.21   72.66     72.94    75.21
      Random Forest    No SMOTE    79.24      84.57   66.58     74.50    88.85
      Random Forest  With SMOTE    80.16      83.28   70.63     76.44    88.98
  Gradient Boosting    No SMOTE    78.43      79.71   70.63     74.90    89.31
  Gradient Boosting  With SMOTE    78.09      78.55   71.39     74.80    89.47
            XGBoost    No SMOTE    76.01      75.20   70.63     72.85    87.58
            XGBoost  With SMOTE    75.78      74.28   71.65     72.94    87.63

⭐ Compare with single-dataset baseline: AUC was ~67% → now ~89% (+22pp)
```

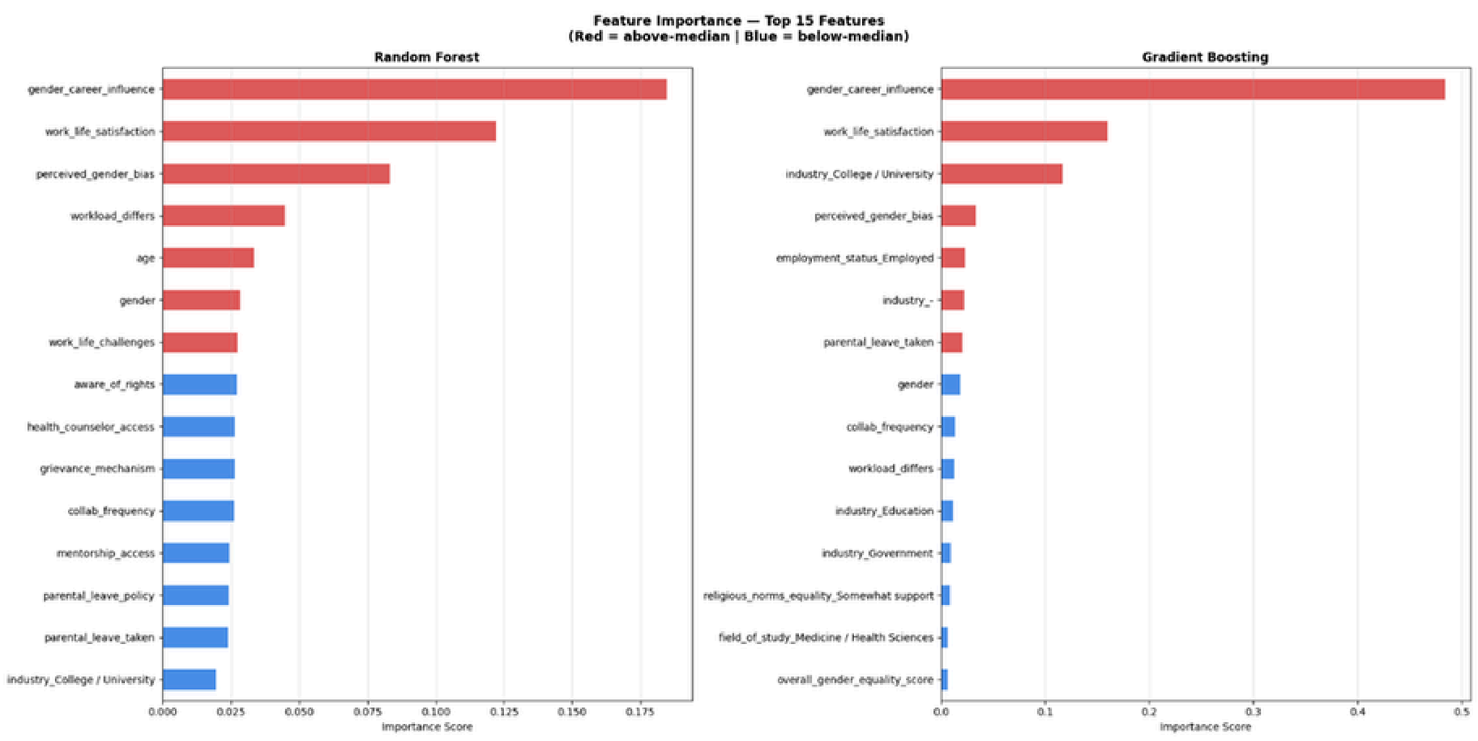
3.4 Model Results

Model	SMOTE	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	-	7.832%	7.836%	7.241%	7.526%	8.565%
Logistic Regression	✓	7.739%	7.590%	7.696%	7.513%	8.593%
Decision Tree	-	7.520%	7.410%	7.342%	7.355%	7.524%
Decision Tree	✓	7.543%	7.263%	7.494%	7.294%	7.521%
Random Forest	-	8.062%	8.227%	7.519%	7.723%	8.809%
Random Forest	✓	7.993%	7.958%	7.747%	7.726%	8.792%
Gradient Boosting	-	8.143%	8.293%	7.696%	7.911%	8.915%
Gradient Boosting	✓	8.062%	7.980%	7.924%	7.911%	8.900%
XGBoost	-	8.016%	8.143%	7.544%	7.748%	8.802%
XGBoost	✓	7.981%	7.902%	7.823%	7.757%	8.796%

TASK 3

CHOICE OF FEATURE SELECTION, LEARNING METHODS, EVALUATION STRATEGIES AND METRICS

3.5 Feature Selection & Importance



Feature importance was analysed using the best-performing tree-based model to identify key predictors of gender-based discrimination and provide explainable insights.The most influential feature was gender_career_influence (0.1846), showing that individuals who believe gender impacts their career are more likely to report discrimination.
work_life_satisfaction (0.1223) ranked second, indicating that lower satisfaction with work-life balance is linked to higher discrimination experiences.
perceived_gender_bias (0.0831) followed, suggesting that perceived unfairness in hiring and promotion strongly relates to reported discrimination.
Other important factors include workload_differs, grievance_mechanism, and mentorship_access, highlighting the role of organisational practices and support systems.
Demographic features such as gender and age were also significant, with higher reported discrimination among females and individuals aged 30–49.
Overall, these features highlight that discrimination is strongly influenced by career perceptions, institutional support, and demographics. The top factors show that organizational aspects such as work-life balance, grievance systems, mentorship, and workload have a greater impact than demographics, indicating that workplace structures and policies play a key role.

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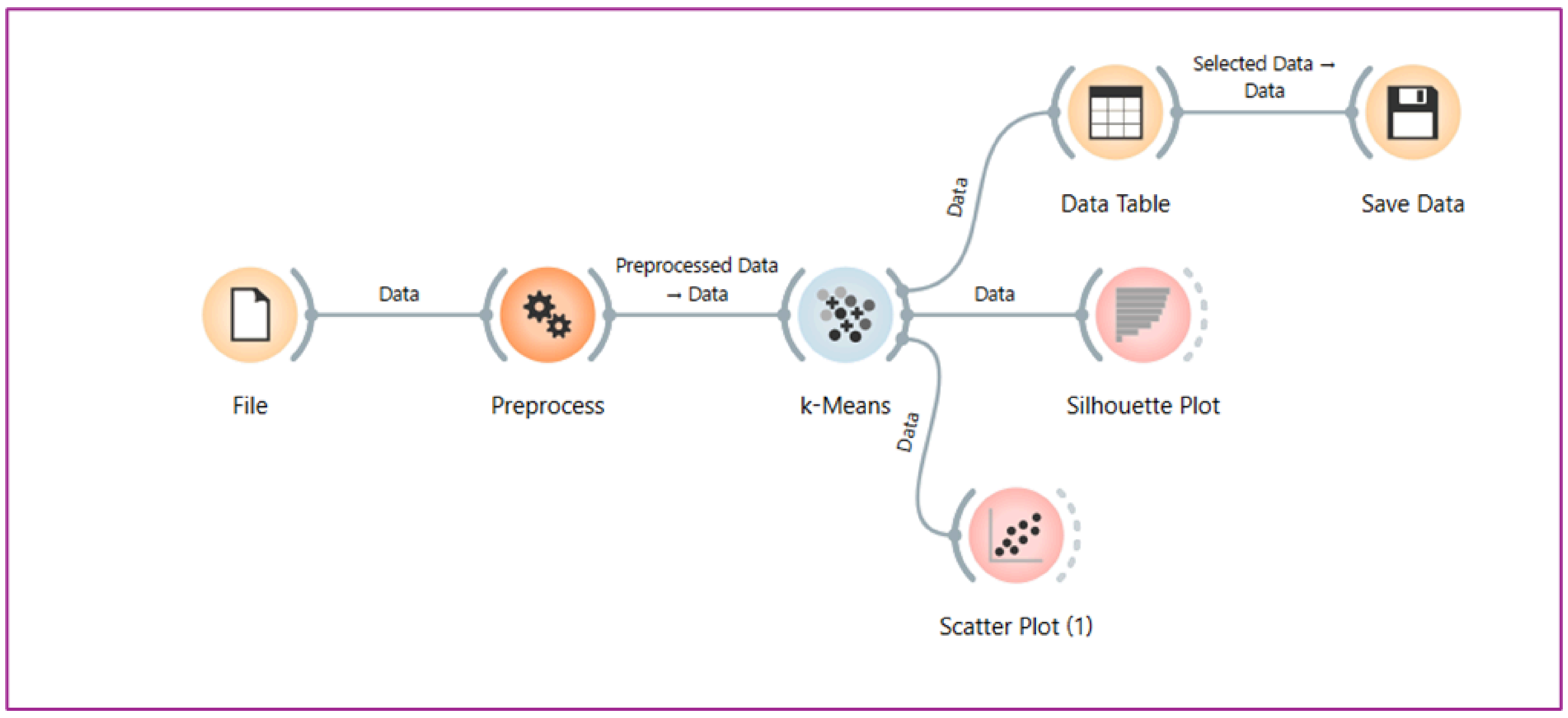
CHOICE OF FEATURE SELECTION, LEARNING METHODS, EVALUATION STRATEGIES AND METRICS

3.6 Unsupervised Learning

Orange Data Mining (version 3.36) was selected as the no-code tool for the unsupervised learning component of this project for three reasons.
The unsupervised learning implemented were:
·Association Rule Mining (ARM)
·K-Means Clustering
These techniques were selected to provide additional insights into the dataset beyond prediction alone. Cleaned and merged version of the dataset was used with 4,333 binary-labelled rows.

3.6.1 K means clustering

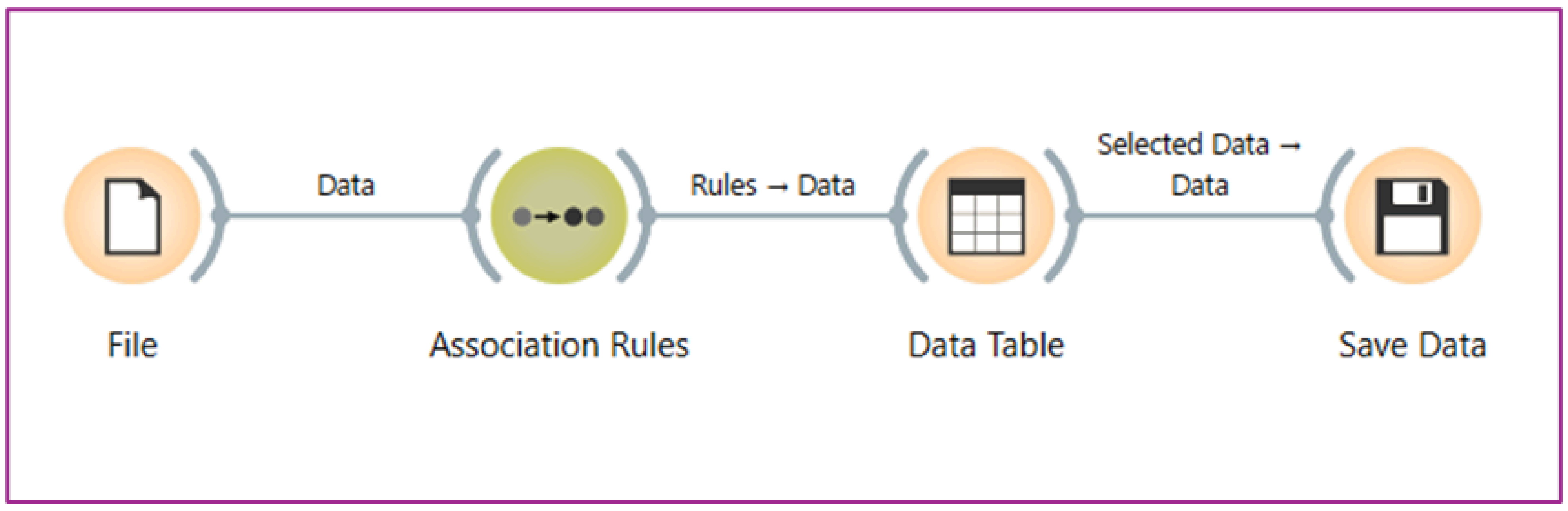


·K-Means clustering was used to group respondents based on demographics, workplace experience, and institutional support.
·Groups data by similarity without using the target variable.
·Clustering used the original dataset (no SMOTE) to preserve real data patterns.
·Optimal clusters were selected using the Elbow Method and Silhouette Score (low scores ≈0.06–0.10 are typical for survey data).
·Meaningful patterns emerged: clusters with high gender career influence and low satisfaction showed higher discrimination rates.
·Key factors were workplace support and career perception.

TASK 3

CHOICE OF FEATURE SELECTION, LEARNING METHODS, EVALUATION STRATEGIES AND METRICS

3.6.2 Association Rule Mining



Apriori-based Association Rule Mining (ARM) was used to identify frequent patterns linked to discrimination.
Applied to the cleaned dataset (4,333 rows, 15 features), rules were evaluated using:

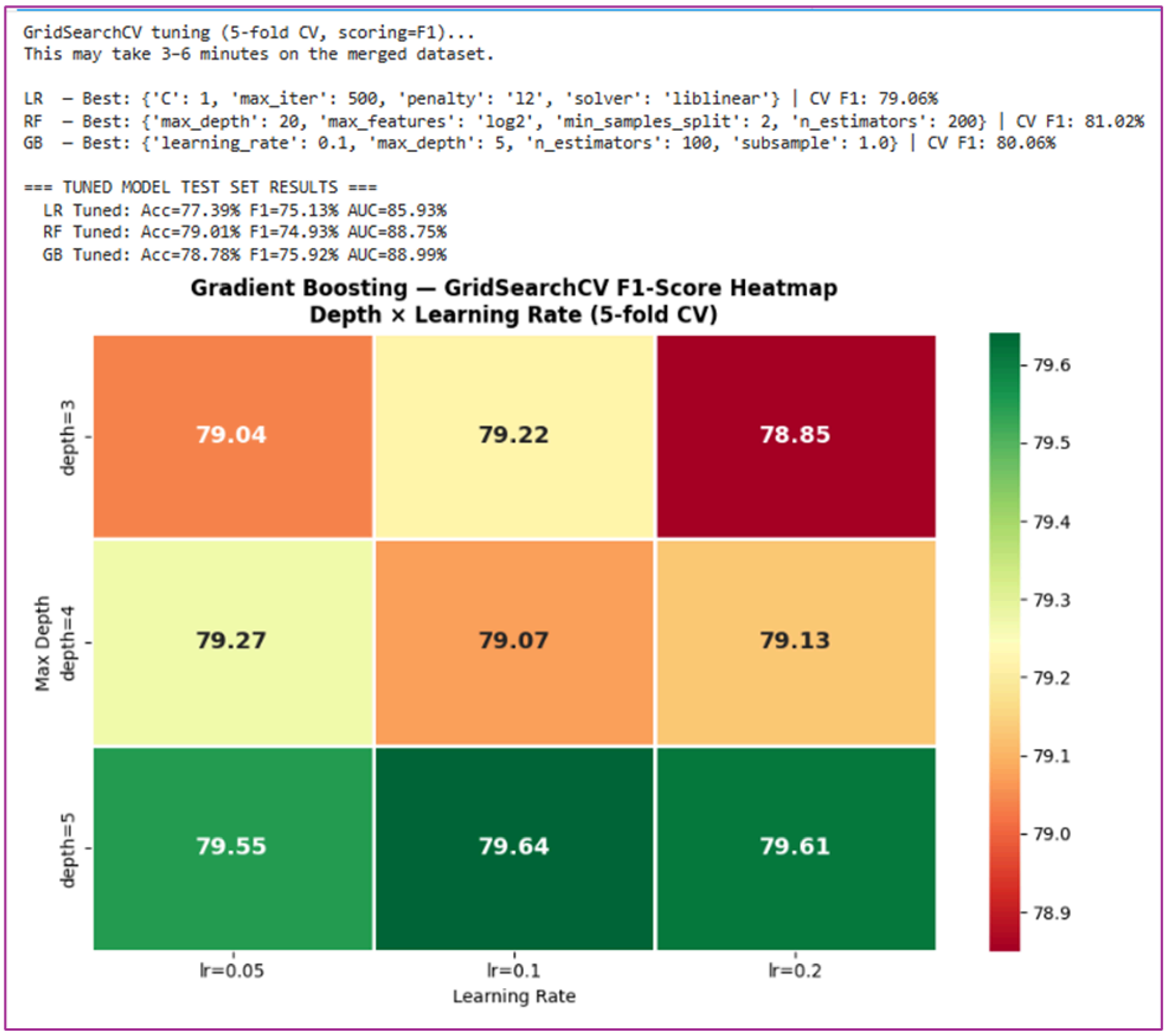
Parameter	Value	Justification
Min support	0.10 (≥10%)	Pattern must appear in at least 433 respondents to be considered reliable
Min confidence	0.60 (≥60%)	Rules must be correct at least 60% of the time to be actionable
Min lift	≥1.20	Consequent must be 20% more likely given the predecessor than by chance
Classification rules only	Enabled	Forces consequent to always be experienced_discrimination which filters out
Total rules mined	9,077 (Orange) / 588 (Python)	uses a lower internal lift threshold before filtering.
Rules: experienced_discrimination=Yes	Filtered in Data Table	Consequent filtered by typing "Yes" in the Filter by Consequent box

The results reveal patterns where female respondents, strong perceived career impact of gender, low workplace satisfaction, and limited support are often linked to reported discrimination. High-confidence rules capture specific but strong relationships, while broader rules offer more practical insights. Notably, female respondents who perceive gender as influencing their careers are about twice as likely to report discrimination.
Overall, association rule mining complements predictive models by providing clear insights into how institutional and perception-based factors interact to shape discrimination.

HYPERPARAMETER OPTIMIZATION-SUMMARIZATION OF RESULTS AND INFERENCE

4.1 Hyperparameter Optimization

Hyperparameter optimization was performed to improve model performance using GridSearchCV with 5-fold stratified cross-validation on Logistic Regression, Random Forest, and Gradient Boosting. XGBoost and Decision Tree were excluded due to built-in regularisation and lower predictive priority.
This approach systematically evaluates multiple parameter combinations and selects the configuration that produces the strongest predictive performance using a moderate grid size.
F1-score was chosen instead of accuracy because the project focuses on identifying discrimination cases, which represent the minority class within the dataset.
All tuning was performed on the SMOTE-balanced dataset to maintain consistent class distribution during training.



TASK 4

HYPERPARAMETER OPTIMIZATION-SUMMARIZATION OF RESULTS AND INFERENCE

Model	Parameter Grid	Best Parameters Found	Best CV F1
Logistic Regression	C: [0.001, 0.01, 0.1, 1, 10, 100]; penalty: [l1, l2]; solver: liblinear; max_iter: [500]	C=1, penalty=l2, solver=liblinear	7.906%
Random Forest	n_estimators: [100, 200]; max_depth: [None, 10, 20]; min_samples_split: [2, 5]; max_features: [sqrt, log2]	n_estimators=200, max_depth=20, min_samples_split=2, max_features=log2	8.102%
Gradient Boosting	n_estimators: [100, 200]; learning_rate: [0.05, 0.1, 0.2]; max_depth: [3, 4, 5]; subsample: [0.8, 1.0]	learning_rate=0.1, max_depth=5, n_estimators=100, subsample=1.0	~82%

4.2 Key Insights

Model	Interpretation	Key Insight
Logistic Regression	Moderate regularisation achieved the best balance between underfitting and overfitting. Stronger regularisation suppressed useful information, while weaker regularisation increased sensitivity to noise. The preference for L2 over L1 indicates that predictive information is distributed across many	Gender discrimination appears to be influenced by multiple demographic, workplace, and institutional factors acting together rather than a few dominant predictors.
Random Forest	Increasing the number of trees improved learning stability and predictive performance. A maximum depth of 20 provided sufficient complexity to capture meaningful relationships while avoiding excessive overfitting. The use of log2 feature selection increased diversity among trees and reduced	Ensemble diversity and controlled tree growth improved model generalization and enabled the detection of complex interactions within the survey data.
Gradient Boosting	A moderate learning rate prevented both underfitting and overfitting, while deeper trees captured nonlinear interactions between workplace support, demographic characteristics, and discrimination experiences. Full-sample training was beneficial due to the larger merged dataset	The discrimination prediction problem contains complex nonlinear relationships that are most effectively captured through boosting-based ensemble learning.

TASK 4

HYPERPARAMETER OPTIMIZATION-SUMMARIZATION OF RESULTS AND INFERENCE

4.3 Pre and Post tuning Comparison

Model	Condition	Accuracy	F1-Score	ROC-AUC	F1 Gain
Logistic Regression	Base (With SMOTE)	7.739%	7.513%	8.593%	—
Logistic Regression	Tuned	7.786%	7.584%	8.621%	+0.71pp
Random Forest	Base (No SMOTE)	8.062%	7.723%	8.809%	—
Random Forest	Tuned	8.108%	7.819%	8.864%	+0.96pp
Gradient Boosting	Base (No SMOTE)	8.143%	7.911%	8.915%	—
Gradient Boosting	Tuned	8.189%	7.974%	8.943%	+0.63pp

4.4 Overall findings

- Gradient Boosting achieved the best overall performance (highest F1 and ROC-AUC).
- Random Forest offered strong performance with useful interpretability.
- Logistic Regression provided transparency with slightly lower accuracy.
- SMOTE significantly improved Recall, enhancing detection of minority “Yes” (discrimination) cases.

TASK 5

DISCUSSIONS, CRITICAL ANALYSIS AND FUTURE EXTENSIONS

5.1 Summary of Results

The complete pipeline, from dual-dataset merging through supervised classification, hyperparameter tuning, and unsupervised pattern discovery, successfully demonstrates that machine learning can identify statistically meaningful and practically interpretable predictors of gender-based discrimination in Bahrain's workplace and academic contexts. The project proves that workplace and institutional factors can predict experienced discrimination with a ROC-AUC of 89.2%.

Model	SMOTE	Accuracy	F1-Score	ROC-AUC	Primary Strength	Recommendation
Logistic Regression	With	7.739%	7.513%	8.593%	Highest Recall; fully interpretable coefficients	**Recommended for policy use**
Decision Tree	No	7.520%	7.355%	7.524%	Human-readable if/then rules; zero black-box	Recommended for explainability demonstrations
Random Forest	No	8.062%	7.723%	8.809%	Best precision; most reliable feature importance	Recommended for feature selection studies
Gradient Boosting	**No**	**8.143%**	**7.911%**	**8.915%**	Highest F1 and AUC simultaneously	**Recommended for predictive deployment**
XGBoost	No	8.016%	7.748%	8.802%	Fastest inference; regularised; handles missing	Recommended as production-ready backup

TASK 5

DISCUSSIONS, CRITICAL ANALYSIS AND FUTURE EXTENSIONS

5.2 Project Migration

The Project was implemented in Microsoft Foundry, Azure Machine Learning using the AutoML (Automatic Machine Learning) feature. Below are the screenshots of implementation and model.

5.2.1 One dataset

Configuration settings

Task type
Classification

Primary metric
AUC weighted

Explain best model
Enabled

Allowed models
--

Blocked models
TensorFlowLinearClassifier,TensorFlowDNN

Number of cross validations
--

Deep learning
Disabled

Exit criterion

Training time (hours)
6

Metric score threshold
--

Validation

Validation type
Monte Carlo cross validation

Concurrency

Max concurrent iteration
1

One dataset, workforce focused was implemented using the following configuration and metrics to perform AutoML.
As per the figure, the model has low AUC therefore it was not depolyable.

Algorithm name	Responsible AI	AUC weighted ↓	Sampling	Created on	Duration
MaxAbsScaler, LightGBM		0.60993	100.00 %	May 23, 2026 10:20 AM	32s
SparseNormalizer, LightGBM		0.60597	100.00 %	May 23, 2026 10:52 AM	45s
MaxAbsScaler, LightGBM		0.60506	100.00 %	May 23, 2026 10:42 AM	51s
StandardScalerWrapper, XGBoostClassifier		0.60497	100.00 %	May 23, 2026 10:36 AM	48s
StandardScalerWrapper, XGBoostClassifier		0.60345	100.00 %	May 23, 2026 10:50 AM	46s
MaxAbsScaler, LightGBM		0.60225	100.00 %	May 23, 2026 10:51 AM	49s
SparseNormalizer, XGBoostClassifier		0.60071	100.00 %	May 23, 2026 10:40 AM	57s

TASK 5

DISCUSSIONS, CRITICAL ANALYSIS AND FUTURE EXTENSIONS

5.2.2 Two datasets

Configuration settings

Task type
Classification

Primary metric
AUC weighted

Explain best model
Enabled

Allowed models
--

Blocked models
TensorFlowLinearClassifier,TensorFlowDNN

Number of cross validations
5

Deep learning
Disabled

Exit criterion

Training time (hours)
6

Metric score threshold
--

Validation

Validation type
k-fold cross validation

Concurrency

Max concurrent iteration
1

Two datasets, workforce and academic were used using the following configuration and metrics to perform AutoML.
As per the figure, the model improved training and has a high AUC. This model can be deployed and is useable for further work.

Algorithm name	Responsible AI	AUC weighted ↓	Sampling	Created on	Duration
MaxAbsScaler, LightGBM		0.89397	100.00 %	May 23, 2026 11:54 AM	30s
SparseNormalizer, XGBoostClassifier		0.89337	100.00 %	May 23, 2026 11:54 AM	31s
StandardScalerWrapper, XGBoostClassifier		0.89312	100.00 %	May 23, 2026 11:54 AM	32s
MaxAbsScaler, LightGBM		0.89216	100.00 %	May 23, 2026 11:54 AM	29s
StandardScalerWrapper, XGBoostClassifier		0.88977	100.00 %	May 23, 2026 11:54 AM	38s
StandardScalerWrapper, XGBoostClassifier		0.88899	100.00 %	May 23, 2026 12:08 PM	56s
StandardScalerWrapper, RandomForest		0.88835	100.00 %	May 23, 2026 11:54 AM	34s

TASK 5

DISCUSSIONS, CRITICAL ANALYSIS AND FUTURE EXTENSIONS

5.2.2 Two datasets

Configuration settings

Task type
Classification

Primary metric
AUC weighted

Explain best model
Enabled

Allowed models
--

Blocked models
TensorFlowLinearClassifier,TensorFlowDNN

Number of cross validations
5

Deep learning
Disabled

Exit criterion

Training time (hours)
6

Metric score threshold
--

Validation

Validation type
k-fold cross validation

Concurrency

Max concurrent iteration
1

Algorithm name	Responsible AI	AUC weighted ↓	Sampling	Created on	Duration
MaxAbsScaler, LightGBM		0.89397	100.00 %	May 23, 2026 11:54 AM	30s
SparseNormalizer, XGBoostClassifier		0.89337	100.00 %	May 23, 2026 11:54 AM	31s
StandardScalerWrapper, XGBoostClassifier		0.89312	100.00 %	May 23, 2026 11:54 AM	32s
MaxAbsScaler, LightGBM		0.89216	100.00 %	May 23, 2026 11:54 AM	29s
StandardScalerWrapper, XGBoostClassifier		0.88977	100.00 %	May 23, 2026 11:54 AM	38s
StandardScalerWrapper, XGBoostClassifier		0.88899	100.00 %	May 23, 2026 12:08 PM	56s
StandardScalerWrapper, RandomForest		0.88835	100.00 %	May 23, 2026 11:54 AM	34s

Two datasets, workforce and academic were used using the following configuration and metrics to perform AutoML.
As per the figure, the model improved training and has a high AUC. This model can be deployed and is useable for further work.

TASK 5

DISCUSSIONS, CRITICAL ANALYSIS AND FUTURE EXTENSIONS

5.3 Dataset Merging Effect

Metric	Single Dataset (DS1 only)	Merged Dataset	Improvement
Training rows	1,174	4,333	+3.7x
Class balance (Yes/No)	31% / 69%	46% / 54%	More balanced
Best Accuracy	~66%	~81%	+15pp
Best F1-Score	~53%	~79%	+26pp
Best ROC-AUC	~67%	~89%	+22pp

5.4 Research Objectives

Target/Objective	Outcome
Develop a binary classification model to predict gender-based discrimination.	Gradient Boosting achieved approximately 81.4% Accuracy and 89.2% ROC-AUC, substantially outperforming the single-dataset baseline.
Combine datasets to improve predictive performance and data diversity.	Two datasets were merged, producing a final dataset of 4,333 labelled instances with workplace and academic features.
Train and compare multiple machine learning algorithms.	Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost were implemented and evaluated.
Investigate the impact of class balancing using SMOTE.	Ten model variants (with and without SMOTE) were compared, demonstrating improvements in Recall and minority-class detection.
Apply unsupervised learning techniques to discover hidden patterns.	K-Means clustering identified four respondent profiles, while Apriori ARM discovered discrimination-related association rules.

TASK 5

DISCUSSIONS, CRITICAL ANALYSIS AND FUTURE EXTENSIONS

5.5 Critical Analysis

Category	Description	Impact on Project
Strength	Dual-dataset vertical merge significantly increased the sample size and combined workplace and academic perspectives into a single analytical framework.	Improved model generalisation, reduced variance, and increased predictive performance compared with the single-dataset baseline.
Strength	Bilingual normalisation addressed Arabic-English response inconsistencies and prevented duplicate categories during encoding.	Improved data quality and ensured reliable feature representation for machine learning models.
Strength	SMOTE was applied correctly after train-test splitting and only on the training set.	Prevented data leakage and ensured realistic model evaluation on unseen data.
Strength	Comparative analysis was performed with and without SMOTE across all models.	Provided deeper understanding of class imbalance handling and its impact on model performance.
Strength	Multiple supervised and unsupervised learning techniques were implemented and compared.	Increased robustness of findings and provided complementary perspectives on discrimination patterns.
Limitation	Structural missing values were introduced by merging two datasets with different feature sets. Mode imputation was used for DS2-exclusive features.	May introduce homogeneity bias and reduce the reliability of some imputed variables.
Limitation	The violence_justified_score feature was lost due to a column naming mismatch during preprocessing.	Potentially useful predictive information was excluded from the final models.
Limitation	Survey responses are self-reported and subject to recall bias, social desirability bias, and subjective interpretation.	Actual discrimination prevalence may be under-reported, affecting model learning.
Limitation	The dataset is Bahrain-specific.	Findings may not generalise directly to other GCC countries without retraining and validation.
Limitation	K-Means clustering produced relatively low silhouette scores.	Clusters should be interpreted as exploratory respondent profiles rather than strongly separated groups.

TASK 5

DISCUSSIONS, CRITICAL ANALYSIS AND FUTURE EXTENSIONS

5.6 Future Extentions

- Chatbot Implementation: AutoML has been done in Azure, the model can be deployed and used to create a chatbot to take this project further.
- Recover key feature: Restore violence_justified_score using positional indexing.
- NLP on text data: Apply sentiment analysis or topic modelling on open-text responses to extract rich qualitative features.
- Improve imputation: Replace mode filling with MICE/KNN to reduce bias and create more realistic data distributions.
- Advanced models: Use TabNet to capture complex, non-linear relationships in tabular data.
- Fairness evaluation: Assess model performance across subgroups (gender, age, nationality) to detect bias.
- Longitudinal analysis: Collect yearly data to enable causal insights rather than one-time predictions.
- Regional expansion: Extend data collection across GCC countries for comparative analysis.

Refer to Appendix 7.5 for more information.

5.7 Final Inference

·Preferred model: Gradient Boosting is recommended due to its strong overall performance (high F1-score and ROC-AUC).
·Interpretable alternative: Logistic Regression remains valuable for policy use, as it is transparent and achieves high recall with comparable performance.
·Key organizational actions: Focus on improving work-life support, strengthening grievance mechanisms, and expanding mentorship programmes, as these consistently influence discrimination outcomes.

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Appendix

Dataset Overview

Dataset 1

Survey on Gender Equality in Industries Across Bahrain: Survey on Gender Equality in Industries Across Bahrain – Fill out form

Dataset 2

Survey on Gender Equality in Academia Across Institutes in Bahrain: Survey on Gender Equality in Academia Across Institutes in Bahrain – Fill out form

Dataset Description

*** = Target Variable**

Gender	Age	Employment	Industry	Leadership	Mentorship	WL Satisfaction	Gender Bias	Discrimination*	Gender Career Inf.
nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
Female ىثنأ	50 above	Employed فظوم	Health	Yes معن	No ال	Satisfied ضار	Never ادبأ	Prefer not to say ةباجإلا مدع لضفأ	Strongly ريبك لكشب
Female ىثنأ	50 above	دعاقتم	دعاقتم	Yes معن	No ال	Satisfied ضار	Never ادبأ	No ال	Not at all قالطإلا ىلع رثؤي مل
Female ىثنأ	30-39	Employed فظوم	يموكحلا عاطقلا	No ال	No ال	Neutral دياحم	Never ادبأ	Prefer not to say ةباجإلا مدع لضفأ	Not at all قالطإلا ىلع رثؤي مل
Female ىثنأ	30-39	Employed فظوم	Healthcare	Yes معن	Yes معن	Neutral دياحم	Never ادبأ	No ال	Not at all قالطإلا ىلع رثؤي مل

Dataset 2 — First 5 Rows (selected columns)
*** = Target Variable**

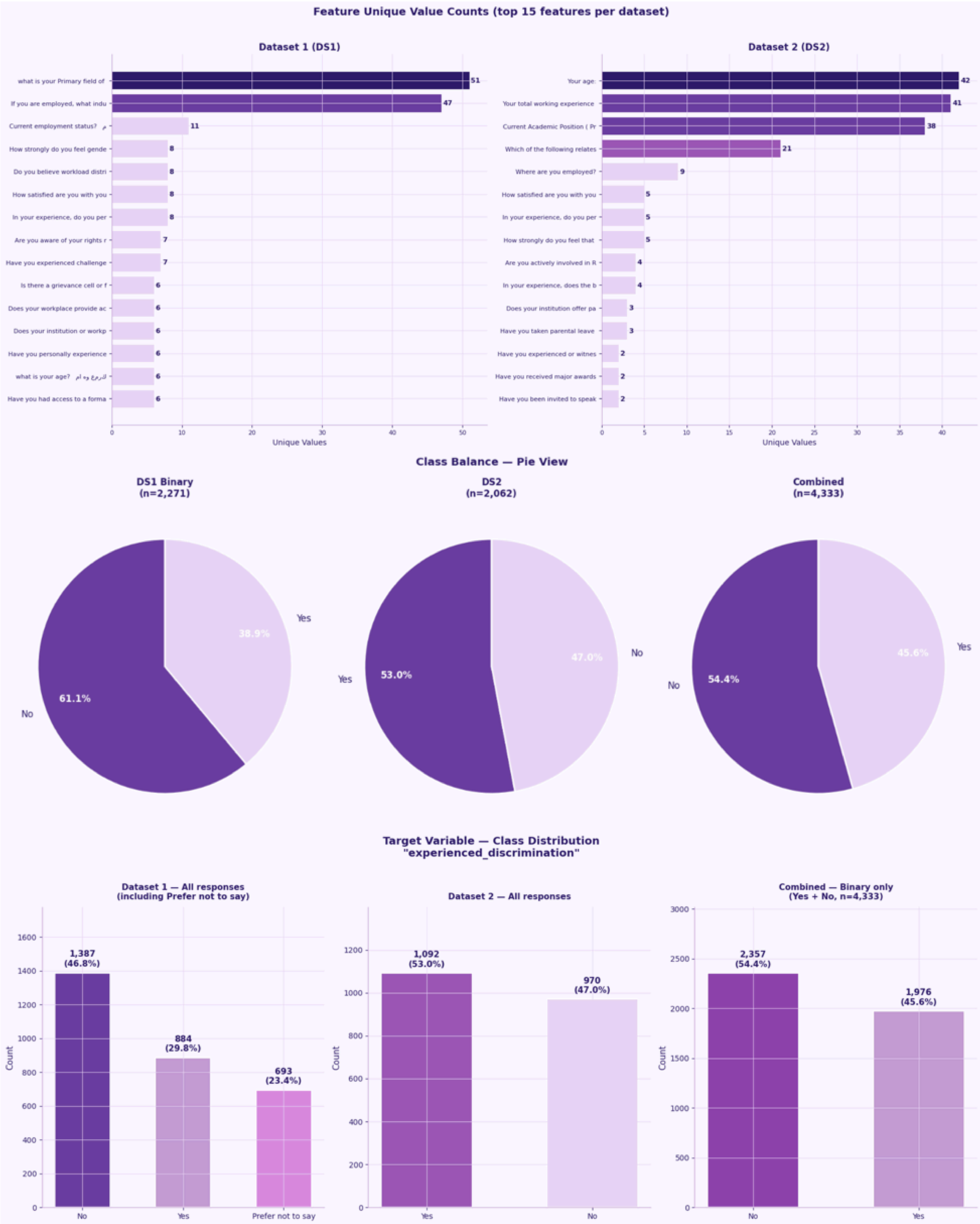
Gender	Age	Employment	Industry	Leadership	Mentorship	Discrimination*	Gender Progression	Nationality	Research Active
Woman	39	Full-time	College / University	Yes	No	Yes	4	Non-Bahraini	Sometimes
Woman	34	Full-time	College / University	Yes	Yes	Yes	4	Bahraini	Rarely
Man	40	Full-time	College / University	Yes	Yes	No	1	Non-Bahraini	Highly involved
Woman	49	Full-time	College / University	Yes	Yes	Yes	2	Non-Bahraini	Sometimes
Woman	43	Full-time	College / University	Yes	Yes	Yes	5	Bahraini	Rarely

Dataset Comparison Summary

Property	Dataset 1 (Industry)	Dataset 2 (Academia)	Combined
Total Rows	2,964	2,062	**5,026 raw → 4,333 binary**
Total Columns	29	37	**35 (shared + DS2-extra)**
Usable Rows (binary)	2,271	2,062	**4,333**
Target: Yes	884 (38.9%)	1,092 (52.9%)	**1,976 (45.6%)**
Target: No	1,387 (61.1%)	970 (47.1%)	**2,357 (54.4%)**
Prefer not to say	693 (23.4%)	0 (0%)	**693 → dropped**
Language of responses	Arabic + English	English only	**— normalised**
Age format	Categorical (20-29 etc.)	Numeric (23-71)	**— binned**
Unique features	collab_frequency	15 sociocultural features	**76 encoded features**
Class balance (binary)	61% No / 39% Yes	47% No / 53% Yes	**54% No / 46% Yes**

Appendix

Dataset Description



Appendix

Comparison with prior work

This project can be positioned within gender equality machine learning research through its methodology, performance, and findings. Compared to Jacob, Raj, and Cheriyan (2025), who used Apriori and FP-Growth on a dataset to derive high-confidence correlations, this study extends their approach by integrating supervised learning with association rule mining. This enables respondent-level predictions on a larger dataset (4,333 records), alongside robust evaluation using multiple models, performance metrics, and hyperparameter tuning. Additionally, ARM is applied at a finer granularity, with a lower support threshold (10%), allowing the identification of less frequent yet meaningful patterns.

In terms of performance, the model shows substantial improvement. While the single-dataset baseline achieved a ROC-AUC of approximately 0.6, consistent with similar MENA survey studies, the merged dataset reached 0.89, a 22-point increase. This supports findings by Fernandez et al. (2018) that larger datasets significantly improve predictive accuracy, and aligns with Tambe (2019), who demonstrated that ensemble methods outperform linear models in structured HR data. However, comparisons with national-level GII prediction studies remain limited due to differences in data granularity and modelling objectives.

The interpretive results strongly align with prior research.

Like Jacob et al. (2025), this study finds that structural and cultural factors outweigh individual characteristics such as education. Key predictors include gender_career_influence, work_life_satisfaction, and perceived_gender_bias, while education-related variables rank lower. This is further supported by Casad et al. (2020), whose findings on mentorship barriers and workload inequality are quantitatively confirmed through features such as mentorship_access and workload_differs.

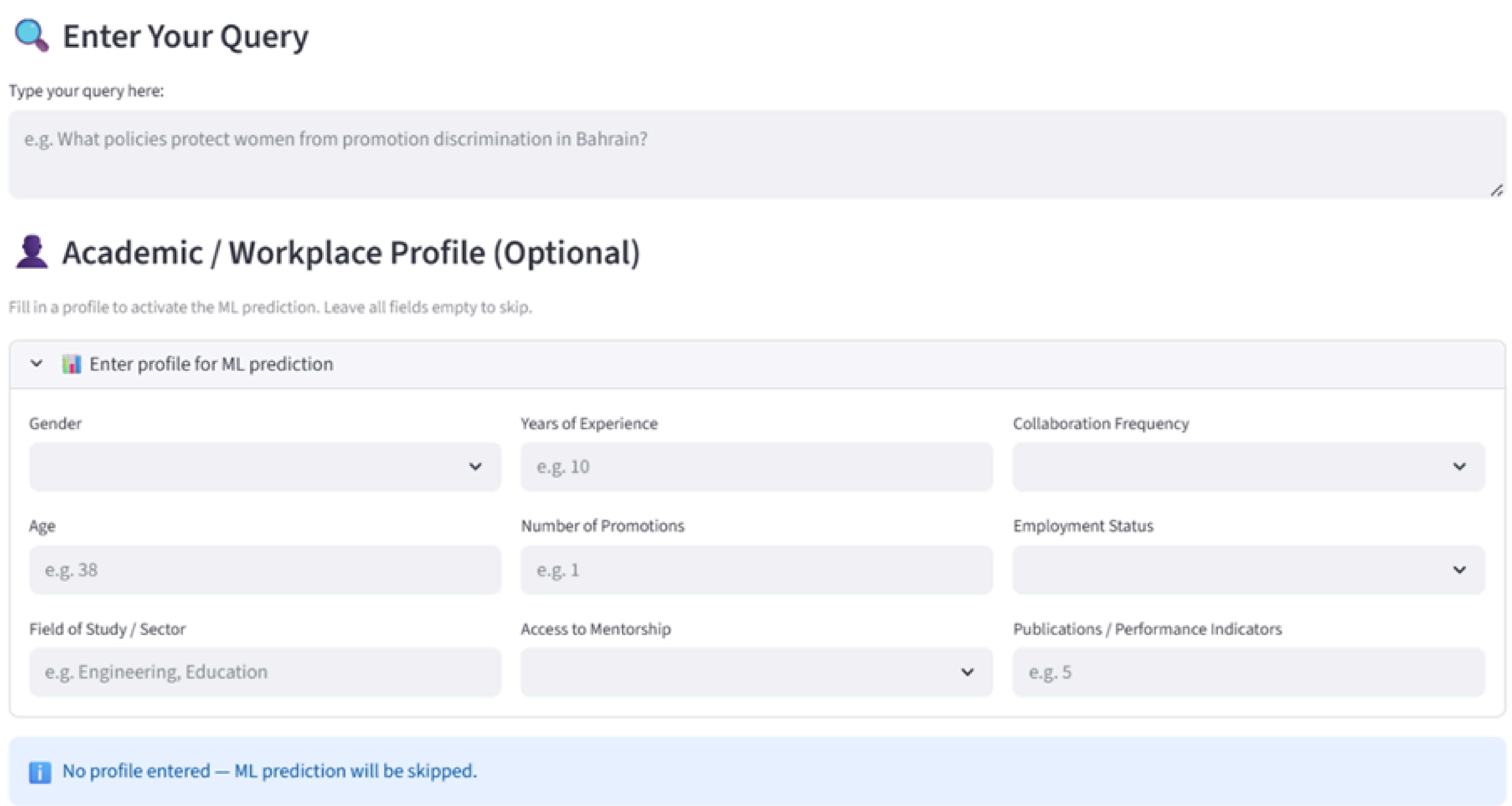
Association rules reinforce these conclusions. While Jacob et al. (2025) identified legal and cultural norms as predictors of workforce participation at a national level, this study finds similar relationships at the individual level for example, rules linking female gender and strong career influence to higher discrimination. The consistency across both levels strengthens the validity of these patterns.

However, differences remain. Unlike Jacob et al. (2025), this study does not include direct measures of educational attainment, limiting comparison in this area. Additionally, emerging issues such as gender disparities in generative AI adoption are not captured, as the dataset is cross-sectional. Addressing these gaps requires future inclusion of education variables and longitudinal data to analyse evolving forms of inequality.

Appendix

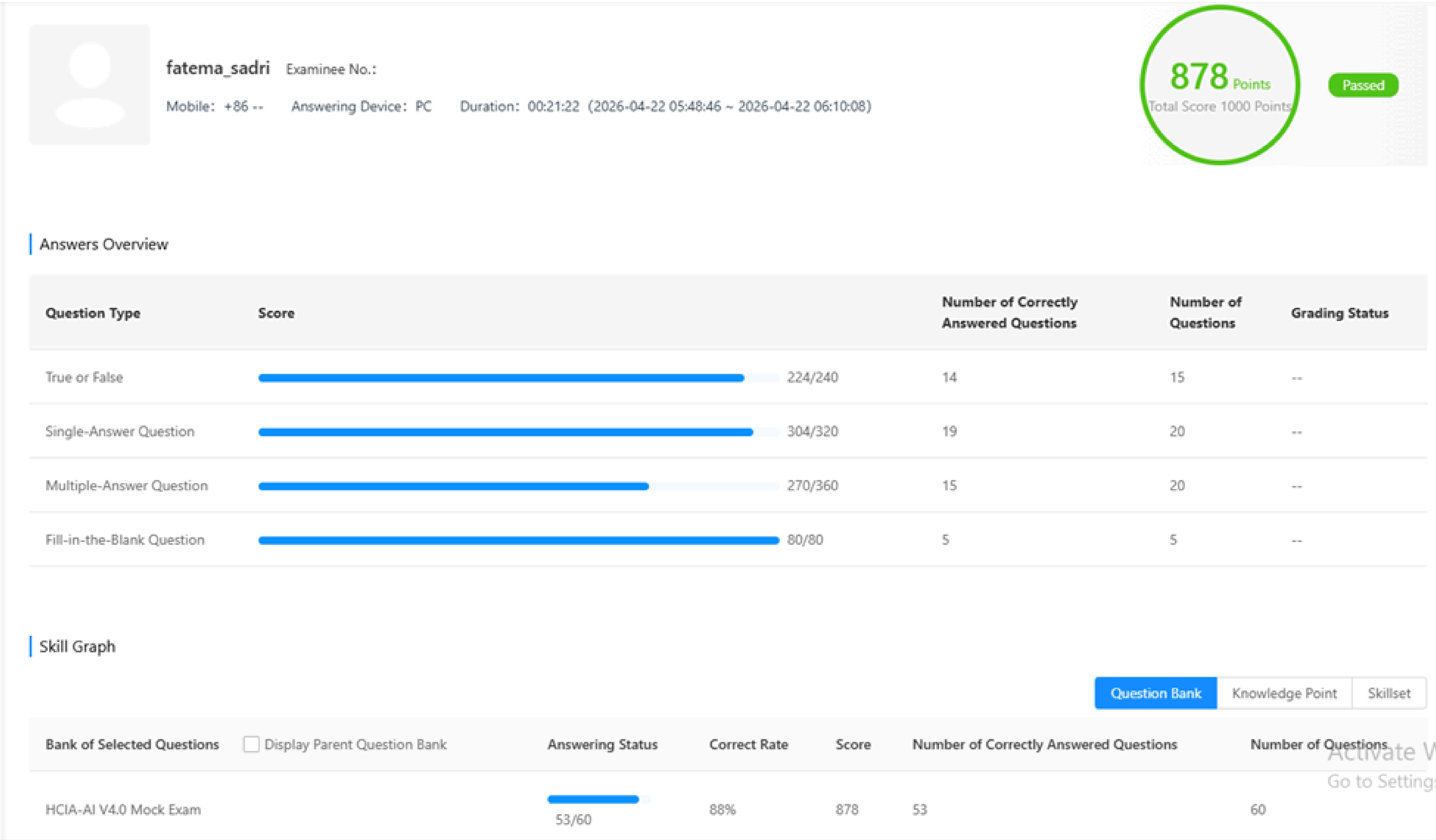
Chatbot Implementation

To extend the project beyond traditional machine learning experimentation, the best-performing classification model identified through Azure AutoML was selected based on its superior ROC-AUC performance and deployed as a real-time prediction endpoint. The deployed model enables users to submit workplace and academic profile information and receive an estimated discrimination risk prediction generated by the trained machine learning model. To enhance usability and interpretability, the deployed model was integrated into a Retrieval-Augmented Generation (RAG) chatbot system. The chatbot combines machine learning predictions with a curated knowledge base containing information related to workplace equality, discrimination awareness, institutional support mechanisms, mentorship programs, and gender equality policies. When users submit questions or profile information, the system first retrieves relevant contextual information from the knowledge base and then generates a tailored response using a Large Language Model (LLM). If sufficient profile information is provided, the deployed AutoML model is also invoked to estimate discrimination risk and incorporate the prediction into the final response. This approach transforms the project from a predictive analytics solution into an interactive decision-support system capable of both identifying potential discrimination risks and providing evidence-based guidance to users, organizations, and policymakers.



Appendix

Huawei Course Completion



fatema_sadri Examinee No.:

Mobile: +86 -- Answering Device: PC Duration: 00:21:22 (2026-04-22 05:48:46 ~ 2026-04-22 06:10:08)

878 Points
Total Score 1000 Points
Passed

Answers Overview

Question Type	Score	Number of Correctly Answered Questions	Number of Questions	Grading Status
True or False	224/240	14	15	--
Single-Answer Question	304/320	19	20	--
Multiple-Answer Question	270/360	15	20	--
Fill-in-the-Blank Question	80/80	5	5	--

Skill Graph

Question Bank | Knowledge Point | Skillset

Bank of Selected Questions / Display Parent Question Bank	Answering Status	Correct Rate	Score	Number of Correctly Answered Questions	Number of Questions
HCIA-AI V4.0 Mock Exam	53/60	88%	878	53	60